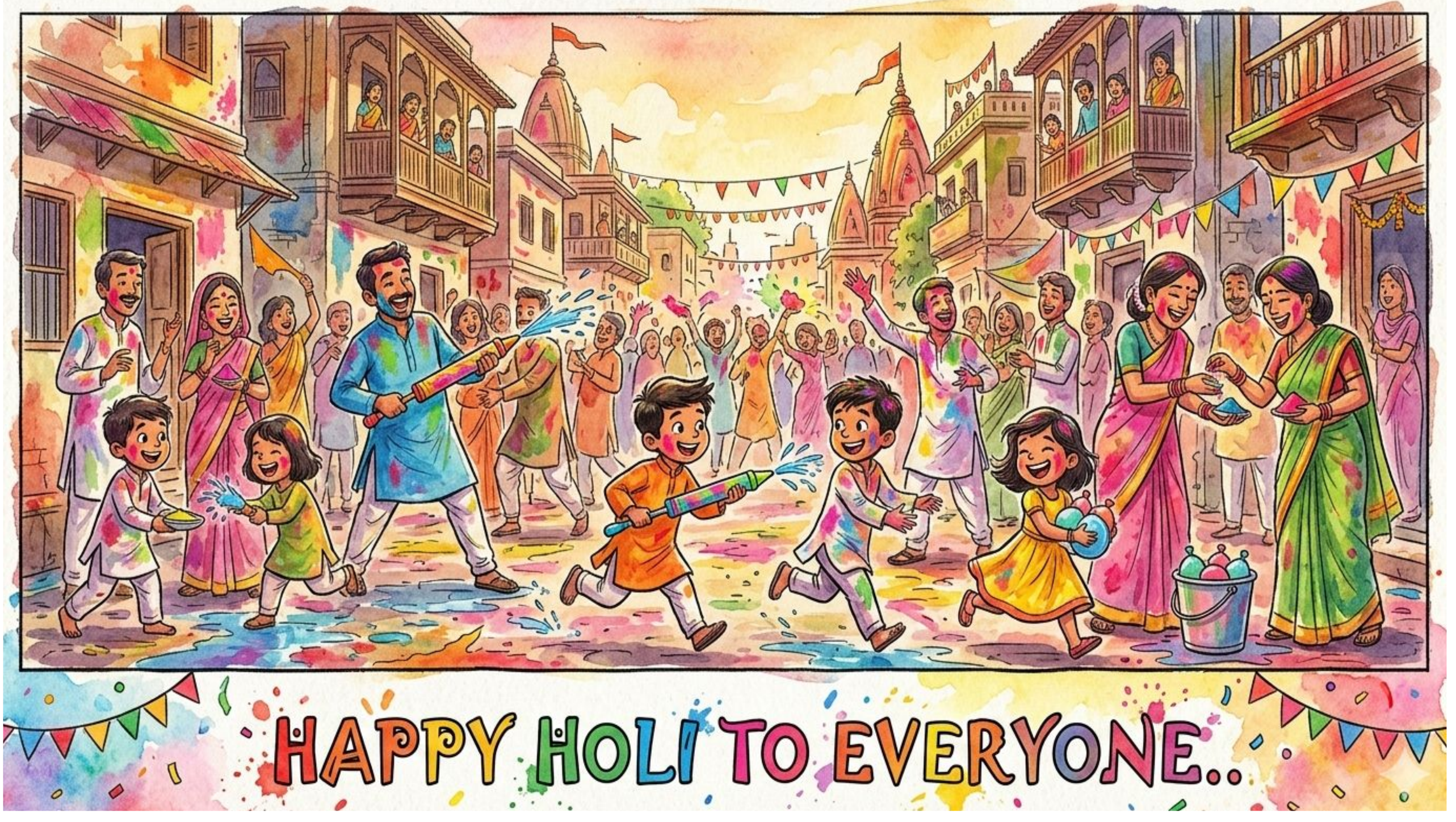
HAPPY HOLI TO EVERYONE..

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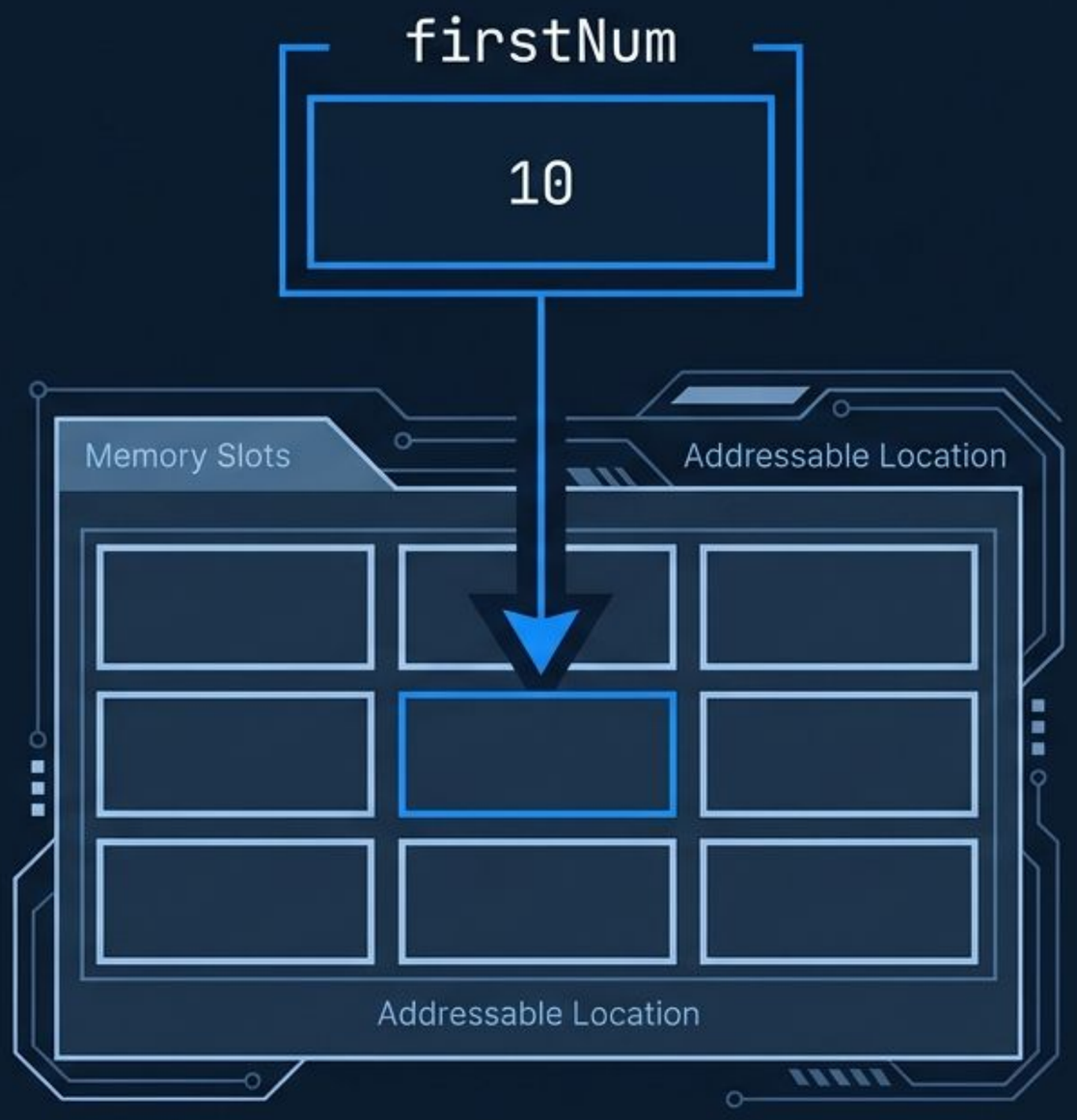
Variables as Data Containers

- Variables act as containers for storing data values.
- Named memory locations for calculations and data reuse.
- **Identifiers** are unique names given to these containers.
- **Literals** are the actual constant values stored inside.



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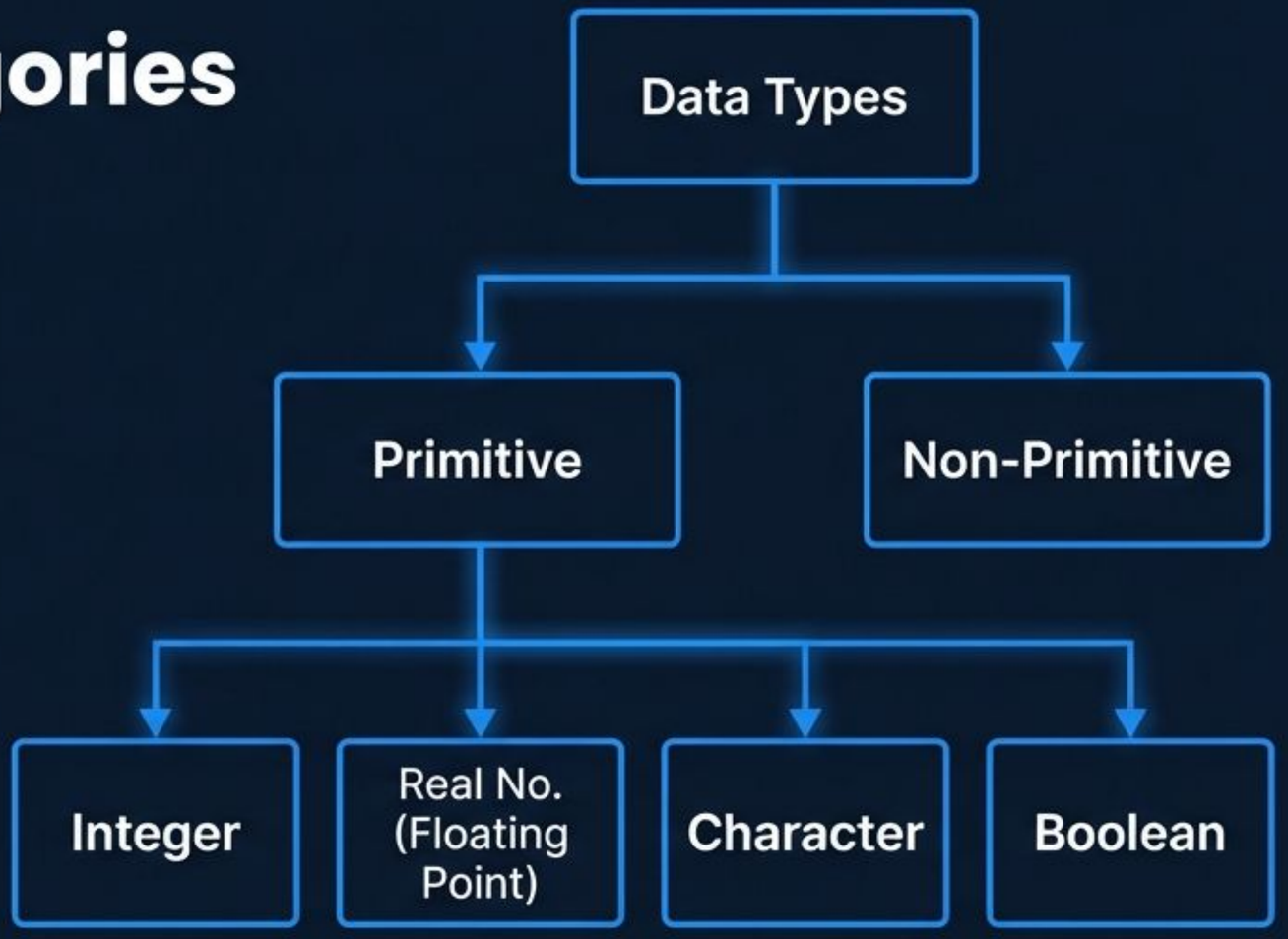
Java Data Type Categories

- Java is statically typed, requiring types before compilation.
- Categorised into Primitive and Non-Primitive data types.
- Primitive types are basic building blocks of data.
- Four categories: Integer, Floating Point, Character, and Boolean.



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Whole Number Representations

- Four types for whole numbers: **byte, short, int, long.**
- **Byte** (8-bit) range is -128 to +127.
- **Int** (32-bit) and **Long** (64-bit) handle larger values.
- Every integer number in Java is **signed** by default.

Type	Width in Bits	Range
`byte`	8	-128 to 127
`short`	16	-32,768 to 32,767
`int`	32	-2.14B to 2.14B
`long`	64	Very large range

Every No. is signed in Java

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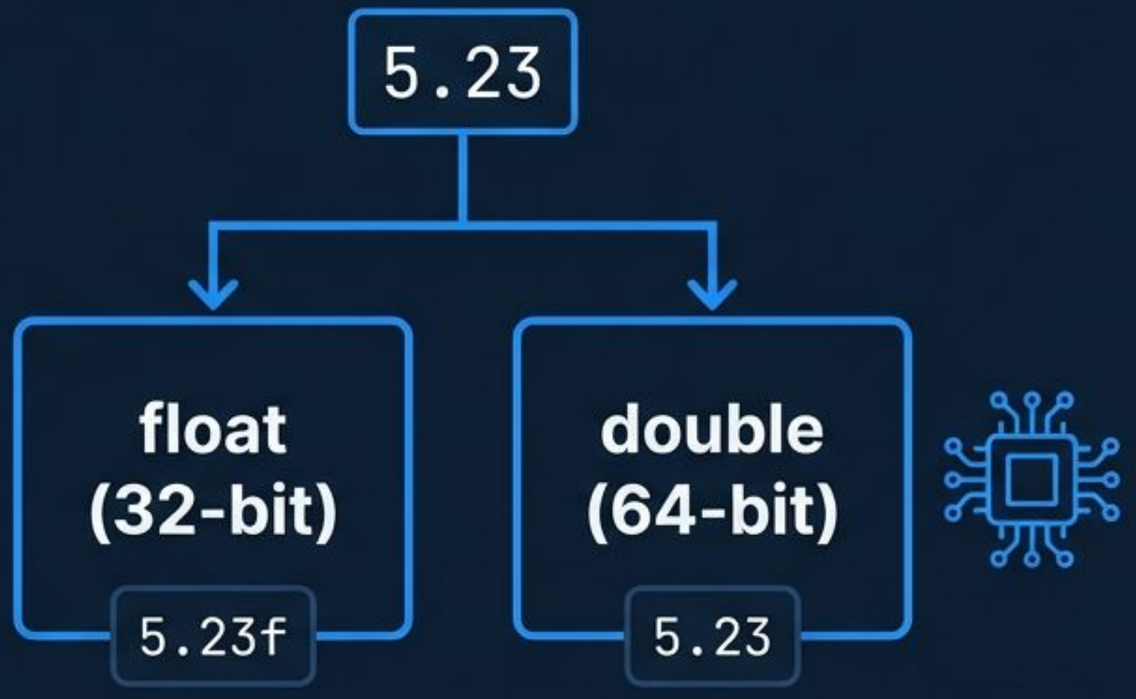
Decimal and Real Numbers

- Used for representing numbers with decimal points.
- Float (32-bit) is known as single-precision.
- Double (64-bit) is double-precision and hardware optimised.
- Requires ‘f’ suffix for float literals specifically.



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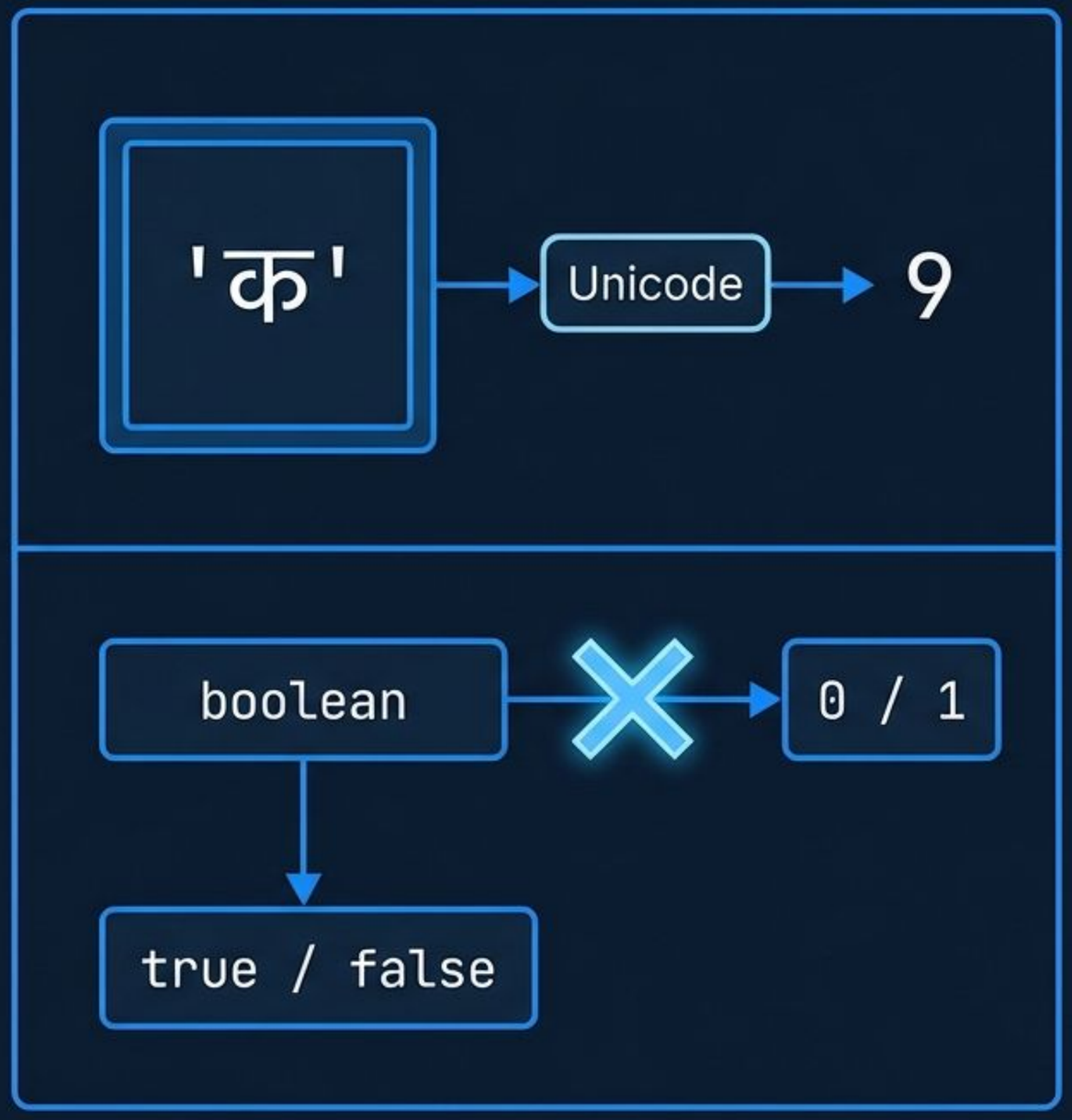
Logical and Unicode Data

- **Char** uses 16-bit Unicode for global language support.
- Represents a single character inside single quotes.
- **Boolean** stores only 'true' or 'false' logical values.
- Unlike C++, integers cannot represent boolean values.



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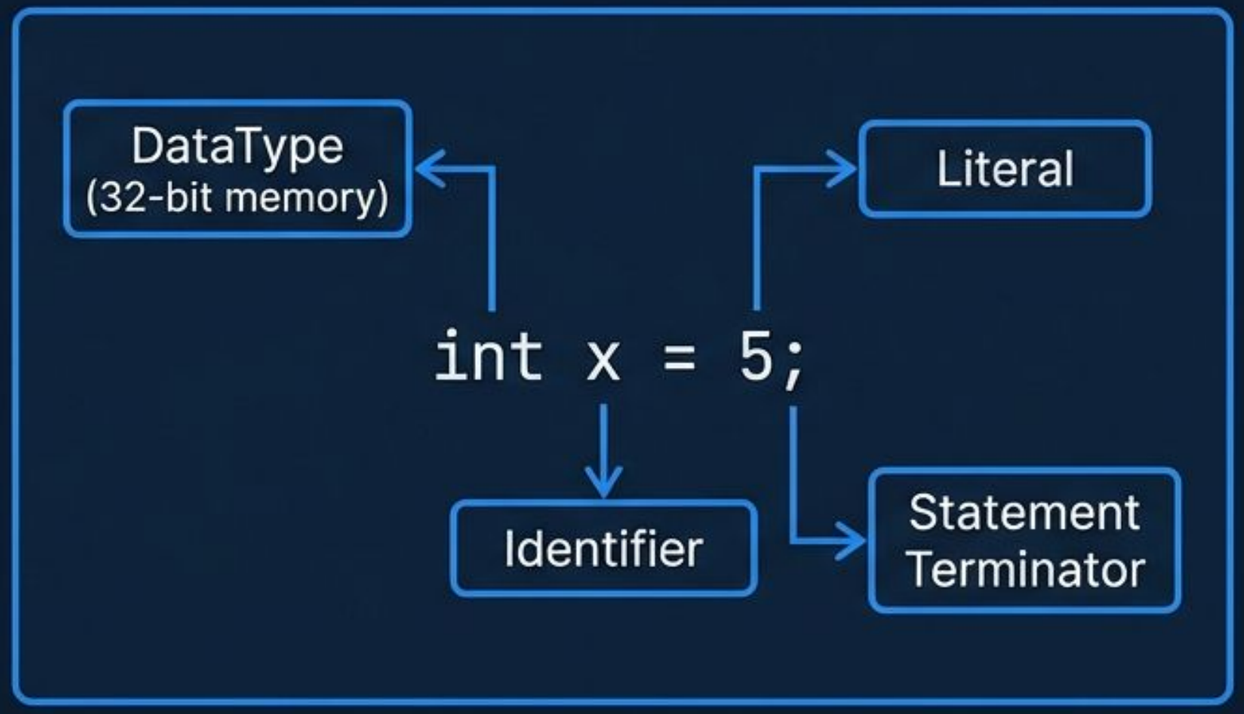
Variable Syntax and Memory

- Declaration creates variable name and specifies its type.
- Definition assigns a specific literal value to it.
- Syntax: DataType Identifier = Value; ends with semicolon.
- Memory is allocated for the container during declaration.



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Reserved Words and Notes

- Keywords are reserved words with special predefined meanings.
- Java has 68 keywords, including 'goto' and 'const'.
- Single-line comments start with double slashes //.
- Multi-line comments use /* text */ syntax for clarity.

public

static

void

class

```
// This is a note

/* Multi-line
   explanation */
```

goto, const